

Shivam Sonawane

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[GitHub](#) | [LinkedIn](#) | [Kaggle](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI/ML Engineer and Data Scientist with hands-on experience building and deploying **machine learning models**, **LangGraph multi-agent systems**, and **data-driven analytics pipelines** in production. Delivered end-to-end solutions across **finance** (SEC data extraction, stock forecasting, sentiment analysis), **QA automation** (self-healing test infrastructure with ML/LLM), and **industrial engineering** (YOLO-based P&ID object detection). Skilled in **statistical modeling**, **feature engineering**, **EDA**, and building **ML inference APIs** using Python, FastAPI, and Docker. Strong foundations in **Natural Language Processing (NLP)**, **deep learning**, prompt engineering, and **MLOps**.

TECHNICAL SKILLS

Languages	Python, SQL
Data Science	Pandas, NumPy, SciPy, Scikit-learn, EDA, Feature Engineering, Statistical Analysis, A/B Testing, Data Preprocessing
Machine Learning	XGBoost, LightGBM, Random Forest, LSTM, Gradient Boosting, Cross-Validation, Hyperparameter Tuning, MLflow, Weights & Biases
Deep Learning	TensorFlow, PyTorch, CNNs, Transformers, YOLO Object Detection, Transfer Learning
GenAI & NLP	LLMs, RAG, LangChain, LangGraph, Ollama, Fine-tuning, Prompt Engineering, Embeddings, Vector Databases (FAISS, ChromaDB, Pinecone), AI Agents, MCP, Sentiment Analysis, OpenAI API, Anthropic API
Visualization	Matplotlib, Seaborn, Plotly, Tableau, Streamlit Dashboards
MLOps & DevOps	Docker, Docker Compose, GitHub Actions CI/CD, Nginx, Celery, Redis, REST APIs
Cloud & Databases	AWS (S3, EC2), Azure ML Studio, PostgreSQL, ChromaDB ,Hugging Face

WORK EXPERIENCE

Gen AI Developer · *Combat Solutions Pvt. Ltd.* 02/2026 – Present

Nashik, India

- Built **production-grade ReAct AI agents** with dynamic tool routing, memory management, and Gmail/API integrations; deployed across **3 client workflows**, reducing manual task handling by an estimated **60%**.
- Designed end-to-end **CI/CD infrastructure**: Docker Compose orchestration, GitHub Actions pipelines, Nginx reverse proxy with SSL, achieving **zero-downtime deployments**.
- Developed a **visual workflow automation platform** (n8n-style) with WhatsApp and Slack nodes, enabling non-technical users to build and trigger multi-step business automation workflows.
- Productionized **YOLO object detection pipelines** for P&ID symbol extraction with OpenCV preprocessing and FastAPI inference APIs; graph-based algorithms export structured XAML/SVG to engineering systems.

AI/ML Developer · *Infomanav Autotropic Cloud Technologies Pvt. Ltd.* 03/2025 – 02/2026

Nashik, India

- Built an **LLM-powered data extraction pipeline** to parse structured financial fields from **SEC Edgar filings** (10-K, 10-Q), replacing manual review and cutting processing time by **80%**.
- Developed a **resume parsing service** using LLMs to extract candidate entities (skills, education, experience) with **92% field-level accuracy**; integrated into the company's internal recruitment workflow.
- Engineered a **stock recommendation engine** combining **fundamental ratios**, **technical indicators** (RSI, MACD, Bollinger Bands), and **NLP-based sentiment scoring** to generate ranked buy/sell signals.
- Built a **time-series forecasting model** using **LSTM** and **Gradient Boosting** with **30+ engineered features**; achieved **12% lower MAE** vs. moving-average baselines on held-out test sets.
- Created an interactive **Streamlit/FastAPI analytics dashboard** visualizing daily user activity across **5 KPIs**, adopted by the analytics team for routine reporting.

Data Science Intern · *AiVariant* 08/2024 – 03/2025

Pune, India

- Cleaned and validated **100,000+ data entries weekly**; built **automated preprocessing pipelines** (missing value imputation, outlier detection, normalization) that lifted dataset accuracy by **15%** on internal benchmarks.
- Performed **exploratory data analysis (EDA)** and **feature engineering** across multiple datasets; built **interactive dashboards** using Pandas, Matplotlib, Seaborn, and **Tableau** for stakeholder reporting.

EDUCATION

Bachelor of Computer Science and Engineering · *Sandip Foundation* 07/2022 – 05/2025

Nashik, India

Diploma in Information Technology · *Amrutvahini Polytechnic* 08/2019 – 07/2022

Sangamner, India

PROJECTS

AI Healer – Self-Healing UI Test Infrastructure Platform [↗](#)

Python, FastAPI, PostgreSQL, ChromaDB, Celery, Redis, XGBoost, LightGBM, Mistral 7B/Qwen2.5-coder 7B, Docker Compose

- Built a **self-healing platform** that auto-detects broken DOM selectors in E2E test suites (Playwright, Selenium, Cypress) and repairs them using a **3-strategy ML engine**: rule-based heuristics, **XGBoost/LightGBM candidate scoring**, and **LLM-powered semantic repair** via local Mistral 7B with Gemini/OpenAI fallback.
- Designed a **3-agent AI pipeline** (Investigator, Healer, Validator) orchestrating DOM diff analysis, ChromaDB **vector similarity retrieval** over chunked HTML embeddings, and automated patch generation using **tree-sitter AST parsing**.
- Engineered a **test failure classification API** categorizing failures across **7 types** (selector, timeout, auth, backend, assertion, DB, syntax) with **evidence-backed root cause analysis** and auto-generated code fixes.
- Deployed via **Docker Compose** with dedicated Celery workers (ingestion, analysis, healing queues), Redis broker, PostgreSQL, and **structured JSON logging**.

PID2DCS & ML Pipeline [↗](#)

Python, YOLO, OpenCV, Scikit-learn, FastAPI, CI/CD

- Developed **ML pipelines for P&ID symbol extraction** using **YOLO object detection** and OpenCV preprocessing; improved detection consistency over rule-based baselines.
- Implemented **scalable model-serving APIs** with FastAPI; designed **graph-based inference algorithms** exporting structured XAML/SVG to downstream engineering systems.

AgentForge – Multi-Agent System with MCP Integration [↗](#)

Python, FastAPI, LangGraph, LangChain, PostgreSQL, Docker Compose, GitHub Actions, Nginx, MCP, VectorDB

- Built a **production-grade multi-agent platform** with **LangGraph ReAct agents**, MCP tool integration, and **multi-tenant architecture** supporting agent creation, workflow automation, and secure data isolation.
- Designed a **visual node-based workflow builder** (n8n-style) with WhatsApp and Slack nodes; automated an **end-to-end RFQ workflow** (vendor parsing, duty computation, email generation) reducing manual quoting time by **70%**.

RAG-Based Document Question Answering System [↗](#)

Python, FastAPI, HuggingFace Transformers, FAISS, Sentence-Transformers

- Developed a **Retrieval-Augmented Generation (RAG)** application using FastAPI to answer queries from private documents with **reduced hallucinations** and source-grounded responses.
- Implemented **semantic vector retrieval** using HuggingFace embeddings and **FAISS** similarity search for accurate document-level question answering.

Deep Learning-Based Object Detection for Industrial Automation [↗](#)

Python, TensorFlow, CNN, K-NN, OpenCV

- Designed a **CNN-based object detection system** for real-time industrial automation; benchmarked against K-NN baselines with improved classification accuracy.
- **Published research paper** in IJMRSET (International Journal of Multidisciplinary Research in Science, Engineering and Technology).

CERTIFICATIONS

- **ExcelR Data Science Certified** | 2025 | Credential ID: 18994/07012025
- **Generative AI Foundations Certified** | 2025 | Credential ID: plOwOFvoWscHe6tK

LANGUAGES

English — Professional Working Proficiency | **Marathi** — Native or Bilingual Proficiency | **Hindi** — Conversational Proficiency